Chapter 2: Basic Topology

Section: Finite, Countable, and Uncountable Sets

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Functions and Mappings

Countability

Unions and Intersections

Countability Results

Summary

Why This Section Matters

- Topology requires a precise language for sets and functions

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- Topology requires a precise language for sets and functions
- Not all infinite sets are the same "size" — we need to make this precise
- Key tools: *1-1 correspondence*, *countability*, *Cantor's diagonal argument*

Functions and Mappings

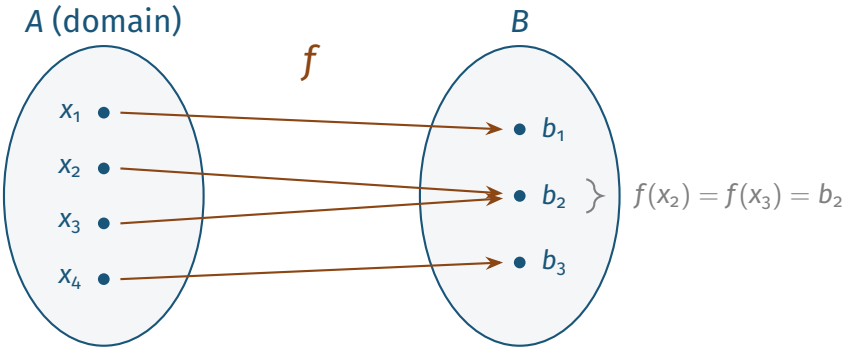
Definition 2.1: Function (Mapping)

Definition 2.1

Consider two sets A and B. If with each element $x \in A$ there is associated an element of B, denoted $f(x)$, then f is a function from A to B (a mapping of A into B).

- **Domain**: the set A
- **Values**: the elements $f(x)$
- **Range**: the set of all values $f(A) = \{f(x) : x \in A\}$

Definition 2.1: Visualizing a Function



Every element of A maps to exactly one element of B

Range $= f(A) = \{b_1, b_2, b_3\} = B$, so this f maps A onto B.

Definition 2.2: Image, Inverse Image, 1-1 Mapping

Definition 2.2 — Image

If $E \subset A$, the image of E under f is:

$$f(E) = \{f(x) : x \in E\}$$

The range of f is $f(A)$. If $f(A) = B$, we say f maps A onto B.

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Definition 2.2 — Inverse Image

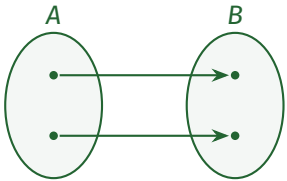
If $E \subset B$, the inverse image of E under f is:

$$f^{-1}(E) = \{x \in A : f(x) \in E\}$$

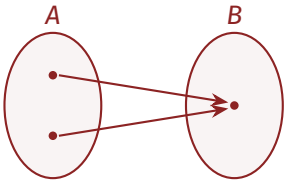
Definition 2.2: Image, Inverse Image, 1-1 Mapping

Definition 2.2 — 1-1 (One-to-One) Mapping

f is 1-1 (one-to-one) if for each $y \in B$, $f^{-1}(y)$ contains at most one element of A. Equivalently: $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$.



1-1



Not 1-1

Definition 2.3: Equivalence of Sets

Definition 2.3 — Equivalence

If there exists a 1-1 mapping of A onto B, we say A and B:

- can be put in 1-1 correspondence
- have the same cardinal number
- are equivalent: $A \sim B$

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This relation is an equivalence relation:

- **Reflexive:** $A \sim A$
- **Symmetric:** $A \sim B \;\Rightarrow\; B \sim A$

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- **Symmetric:** $A \sim B \;\Rightarrow\; B \sim A$
- **Transitive:** $A \sim B$ and $B \sim C \;\Rightarrow\; A \sim C$

Countability

Definition 2.4: Finite, Countable, and Uncountable Sets

Definition 2.4

Let $J_n = \{1, 2, \ldots, n\}$ and $J = \{1, 2, 3, \ldots\}$ (all positive integers).

(a) A is finite	if $A \sim J_n$ for some n (or $A = \varnothing$)
(b) A is infinite	if A is not finite
(c) A is countable	if $A \sim J$
(d) A is uncountable	if A is neither finite nor countable
(e) A is at most countable	if A is finite or countable

Example 2.5: The Integers Are Countable

Example 2.5

The set $A = \mathbb{Z}$ of all integers is countable.

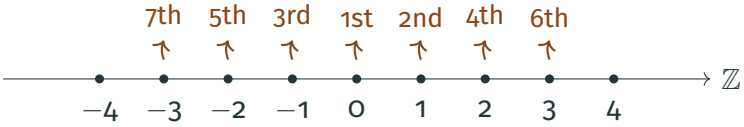
Arrangement:

$$\begin{array}{lrrrrrrrl} A: & 0, & 1, & -1, & 2, & -2, & 3, & -3, & \ldots \\ J: & 1, & 2, & 3, & 4, & 5, & 6, & 7, & \ldots \end{array}$$

Explicit formula:

$$f(n) = \begin{cases} \dfrac{n}{2} & \text{if } n \text{ is even,} \\ -\dfrac{n-1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Example 2.5: Visualizing the Correspondence



Key idea: We "zigzag" through $\mathbb{Z}$, visiting 0, then ± 1, then ± 2, ...

Remark 2.6: Infinite Sets and Proper Subsets

Remark 2.6

A *finite* set cannot be equivalent to any of its proper subsets.

However, an *infinite* set *can* be equivalent to a proper subset of itself!

Example: $J \subset \mathbb{Z}$ is a proper subset, yet $J \sim \mathbb{Z}$ (Example 2.5).

Alternative definition of infinite:
A is infinite $\iff$ A is equivalent to one of its proper subsets.

Definition 2.7: Sequences

Definition 2.7

A sequence is a function f defined on $J = \{1, 2, 3, \dots\}$.

If $f(n) = x_n$, we write $\{x_n\}$ or $x_1, x_2, x_3, \dots$

The values x_n are the terms of the sequence.

Key observations:

- Terms need *not* be distinct
- Every countable set is the range of a sequence of *distinct* terms
- We can "arrange" any countable set in a sequence

Theorem 2.8: Infinite Subsets of Countable Sets

Theorem 2.8

Every infinite subset of a countable set A is countable.

Consequence: Countable sets are the "smallest" infinity.

No uncountable set can be a subset of a countable set.

Proof of Theorem 2.8 — Strategy

Proof Strategy

Given: $E \subset A$, A countable, E infinite.

Goal: Show $E \sim J$ (i.e., E is countable).

Approach:

1. Arrange A in a sequence $\{x_n\}$ of distinct terms.
2. Pick out the elements of E *in order of their indices*.
3. This gives a 1-1 correspondence between E and J.

Proof of Theorem 2.8 — Construction

Step 1: Set up the subsequence

Arrange $A = \{x_1, x_2, x_3, \dots\}$ (distinct terms).

Construct a sequence of indices $\{n_k\}$:

- $n_1 =$ smallest index with $x_{n_1} \in E$
- $n_k =$ smallest index $> n_{k-1}$ with $x_{n_k} \in E$

Step 2: Conclude

Define $f(k) = x_{n_k}$.

Then f is a 1-1 correspondence between J and E. $\square$

Unions and Intersections

Definition 2.9: Union and Intersection

Definition 2.9 — Union

Given a family of sets $\{E_\alpha\}$ indexed by A, the union is:

$$S = \bigcup_{\alpha \in A} E_\alpha = \{x : x \in E_\alpha \text{ for at least one } \alpha \in A\}$$

Special cases: $\bigcup_{m=1}^{n} E_m = E_1 \cup \cdots \cup E_n \qquad \bigcup_{m=1}^{\infty} E_m$

Definition 2.9: Union and Intersection

Definition 2.9 — Union

$$S = \bigcup_{\alpha \in A} E_\alpha = \{x : x \in E_\alpha \text{ for at least one } \alpha \in A\}$$

Definition 2.9 — Intersection

$$P = \bigcap_{\alpha \in A} E_\alpha = \{x : x \in E_\alpha \text{ for every } \alpha \in A\}$$

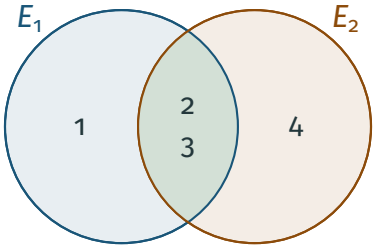
If $A \cap B \neq \varnothing$, we say A and B intersect; otherwise they are disjoint.

Examples 2.10: Union and Intersection

Example 2.10(a)

$E_1 = \{1, 2, 3\}, \quad E_2 = \{2, 3, 4\}$

$E_1 \cup E_2 = \{1, 2, 3, 4\}, \quad E_1 \cap E_2 = \{2, 3\}$



Example 2.10(b): A Continuous Family of Sets

Example 2.10(b)

For each $x \in (0, 1]$, let $E_x = \{y \in \mathbb{R} : 0 < y < x\} = (0, x)$.

(i) $E_x \subset E_z$ if and only if $0 < x \leq z \leq 1$

Example 2.10(b): A Continuous Family of Sets

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(ii) $\displaystyle\bigcup_{x \in (0,1]} E_x = E_1 = (0, 1)$

(iii) $\displaystyle\bigcap_{x \in (0,1]} E_x = \varnothing$

Why? For any $y > 0$, if $x < y$ then $y \notin E_x$. So no y belongs to *every* E_x.

Remarks 2.11: Algebraic Properties of Set Operations

Commutative laws:

$$A \cup B = B \cup A; \qquad A \cap B = B \cap A.$$

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Distributive law:

$$\boxed{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}$$

Remarks 2.11: Algebraic Properties of Set Operations

Further identities (with $\varnothing$ denoting the empty set):

$$A \subset A \cup B, \qquad A \cap B \subset A$$

$$A \cup \varnothing = A, \qquad A \cap \varnothing = \varnothing$$

$$\text{If } A \subset B: \quad A \cup B = B, \quad A \cap B = A$$

Countability Results

Theorem 2.12: Countable Union of Countable Sets

Theorem 2.12

Let $\{E_n\}$, $n = 1, 2, 3, \ldots$, be a sequence of countable sets, and put

$$S = \bigcup_{n=1}^{\infty} E_n.$$

Then S is countable.

In words: A countable union of countable sets is countable.

Proof of Theorem 2.12 — The Diagonal Arrangement

Step 1: Arrange each E_n as a sequence

Write $E_n = \{x_{n1}, x_{n2}, x_{n3}, \dots\}$ and form the array:

$$
\begin{array}{lccccc}
E_1: & x_{11} & x_{12} & x_{13} & x_{14} & \cdots \\
E_2: & x_{21} & x_{22} & x_{23} & x_{24} & \cdots \\
E_3: & x_{31} & x_{32} & x_{33} & x_{34} & \cdots \\
E_4: & x_{41} & x_{42} & x_{43} & x_{44} & \cdots \\
 & \vdots & \vdots & \vdots & \vdots &
\end{array}
$$

diag. 1
diag. 2
diag. 3
diag. 4

Proof of Theorem 2.12 — The Diagonal Sequence

Step 2: Enumerate along diagonals

List the elements in order:

$$x_{11};\ x_{21}, x_{12};\ x_{31}, x_{22}, x_{13};\ x_{41}, x_{32}, x_{23}, x_{14};\ \ldots$$

Step 3: Handle duplicates and conclude

- If sets E_n overlap, some elements repeat $\Rightarrow S \sim T \subset J$
- By Theorem 2.8, S is at most countable
- Since $E_1 \subset S$ and E_1 is infinite, S is infinite
- Therefore S is countable $\square$

Corollary of Theorem 2.12

Corollary

If A is at most countable and each B_α (for $\alpha \in A$) is at most countable, then

$$T = \bigcup_{\alpha \in A} B_\alpha$$

is at most countable.

Proof: T is equivalent to a subset of the array from Theorem 2.12.

Theorem 2.13: Countability of n-Tuples

Theorem 2.13

Let A be a countable set, and let B_n be the set of all n-tuples $(a_1, \ldots, a_n)$ with $a_k \in A$. Then B_n is countable.

Proof of Theorem 2.13

Proof by induction on n

Base case: $B_1 = A$ is countable.

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Inductive step: Assume B_{n-1} is countable.

Elements of B_n have the form (b, a) where $b \in B_{n-1}$, $a \in A$.

Proof of Theorem 2.13

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Base case: $B_1 = A$ is countable.

Inductive step: Assume B_{n-1} is countable.

Elements of B_n have the form (b, a) where $b \in B_{n-1}$, $a \in A$.

For each fixed b, the set $\{(b, a) : a \in A\} \sim A$ is countable.

$\Rightarrow$ B_n is a *countable union of countable sets*.

$\Rightarrow$ B_n is countable by Theorem 2.12. $\square$

Corollary: The Rationals Are Countable

Corollary of Theorem 2.13

The set $\mathbb{Q}$ of all rational numbers is countable.

Proof

Every rational $r = b/a$ where $a, b \in \mathbb{Z}$.

The set of pairs (a, b) is B_2 with $A = \mathbb{Z}$ (countable by Example 2.5).

By Theorem 2.13, B_2 is countable.

Since $\mathbb{Q} \sim T \subset B_2$, $\mathbb{Q}$ is at most countable.

Since $\mathbb{Q}$ is infinite, $\mathbb{Q}$ is countable. $\square$

Theorem 2.14: An Uncountable Set

Theorem 2.14

Let A be the set of all sequences whose elements are the digits 0 and 1. Then A is uncountable.

Elements of A: sequences like $1, 0, 0, 1, 0, 1, 1, 1, \ldots$

Technique: Cantor's diagonal argument

Proof of Theorem 2.14 — Cantor's Diagonal Argument

Step 1: Assume E is a countable subset of A

List the elements of E as sequences $s_1, s_2, s_3, \ldots$:

	1	2	3	4	5	
s_1:	**1**	0	0	1	0	$\cdots$
s_2:	0	**1**	1	0	1	$\cdots$
s_3:	1	0	**0**	0	0	$\cdots$
s_4:	0	1	1	**1**	0	$\cdots$
s_5:	1	1	0	0	**1**	$\cdots$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Proof of Theorem 2.14 — Cantor's Diagonal Argument

Step 2: Construct a new sequence s by flipping the diagonal

- If the nth digit of s_n is 1, set the nth digit of s to 0
- If the nth digit of s_n is 0, set the nth digit of s to 1

From the example:

$$\begin{array}{rcccccc} \text{Diagonal:} & \mathbf{1}, & \mathbf{1}, & \mathbf{0}, & \mathbf{1}, & \mathbf{1}, & \ldots \\ s: & \mathbf{0}, & \mathbf{0}, & \mathbf{1}, & \mathbf{0}, & \mathbf{0}, & \ldots \end{array}$$

s differs from s_n in position n, so $s \neq s_n$ for every n.

Therefore $s \notin E$, yet $s \in A$.

Proof of Theorem 2.14 — Cantor's Diagonal Argument

Step 3: Conclude

Every countable subset E of A is a *proper* subset of A.

If A were countable, then A would be a proper subset of itself — contradiction!

$\boxed{A \text{ is uncountable.}}$

Connection: Binary sequences $\longleftrightarrow$ real numbers (binary representation).

$\Rightarrow$ The set of all real numbers is uncountable.

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- **Uncountable sets exist:** The set of all binary sequences is uncountable (Cantor's diagonal argument, Theorem 2.14)

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- **Countable sets:** $\mathbb{Z}$ and $\mathbb{Q}$ are countable; countable unions of countable sets remain countable (Theorems 2.12, 2.13)
- **Uncountable sets exist:** The set of all binary sequences is uncountable (Cantor's diagonal argument, Theorem 2.14)
- **Set operations:** Union, intersection, and their algebraic properties (Remarks 2.11)

Coming up next: Metric Spaces — adding the concept of *distance* to sets.